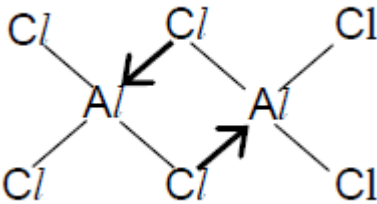
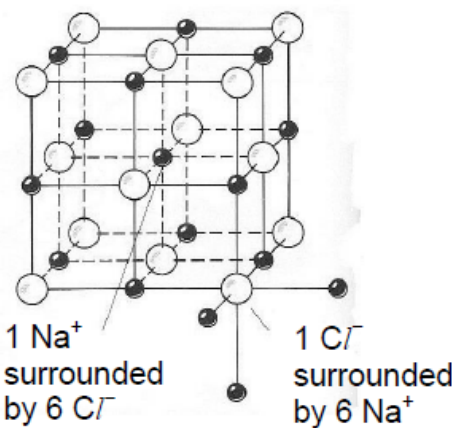
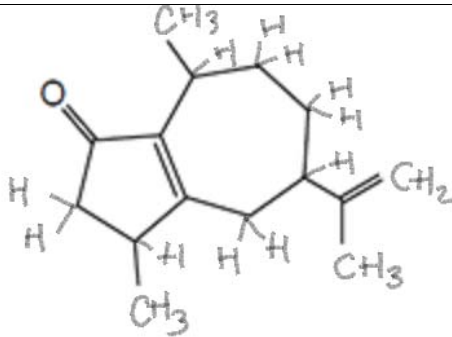


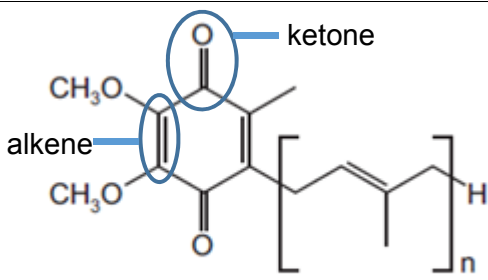
**Yishun Junior College H1 Chemistry 8873 Preliminary Examinations 2018**  
**Paper 1 Solutions**

| 1             | D                        | <p>Option 1 – Incorrect. Nitrogen has a maximum oxidation state of +5.</p> <p>Option 2 - Incorrect. Electronegativity does not explain why nitrogen is unable to expand octet configuration.</p> <p>Option 3 – Correct. Nitrogen is from period 2 and so it does not have energetically accessible d orbitals to expand octet.</p> <p>Hence, only option 3 is correct.</p>                                                                                                                                                                                                                                                                                                                                 |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
|---------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------------|----|-------------|---|----|-------------|---|----|-------------|---|----|-------------|---|----|-------------|---|----|-------------|---|
| 2             | A                        | <p>Total nucleon mass = <math>238 + 2 = 240</math></p> <p>Total proton no = <math>92 + 1 = 93</math></p> <p>Nucleon mass of G = <math>240 - 2 = 238</math></p> <p>Proton no = 93</p> <p>From the Periodic Table, Np has a proton number of 93. Hence G is <math>^{238}_{93}\text{Np}</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                           |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 3             | B                        | <p>Option A – Oxidation number of N do not change while oxidation number of C changes.</p> <p>Option B – Bond angle in <math>\text{NH}_4^+</math>, <math>\text{NH}_3</math>, <math>\text{C}_2\text{H}_4</math> and <math>\text{C}_2\text{H}_6</math> are <math>109.5^\circ</math>, <math>107^\circ</math>, <math>120^\circ</math> and <math>109.5^\circ</math> respectively.</p> <p>Option C - There are no lone pairs of electrons in <math>\text{C}_2\text{H}_6</math>.</p> <p>Option D – There are no <math>\pi</math> bonds in the conversion of <math>\text{NH}_4^+ \rightarrow \text{NH}_3</math>.</p>                                                                                               |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 4             | D                        | <table border="1"> <thead> <tr> <th>proton number</th><th>electronic configuration</th><th>no. of unpaired electrons</th></tr> </thead> <tbody> <tr> <td>13</td><td><math>3s^2 3p^1</math></td><td>1</td></tr> <tr> <td>14</td><td><math>3s^2 3p^2</math></td><td>2</td></tr> <tr> <td>15</td><td><math>3s^2 3p^3</math></td><td>3</td></tr> <tr> <td>16</td><td><math>3s^2 3p^4</math></td><td>2</td></tr> <tr> <td>17</td><td><math>3s^2 3p^5</math></td><td>1</td></tr> <tr> <td>18</td><td><math>3s^2 3p^6</math></td><td>0</td></tr> </tbody> </table>                                                                                                                                                | proton number | electronic configuration | no. of unpaired electrons | 13 | $3s^2 3p^1$ | 1 | 14 | $3s^2 3p^2$ | 2 | 15 | $3s^2 3p^3$ | 3 | 16 | $3s^2 3p^4$ | 2 | 17 | $3s^2 3p^5$ | 1 | 18 | $3s^2 3p^6$ | 0 |
| proton number | electronic configuration | no. of unpaired electrons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 13            | $3s^2 3p^1$              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 14            | $3s^2 3p^2$              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 15            | $3s^2 3p^3$              | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 16            | $3s^2 3p^4$              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 17            | $3s^2 3p^5$              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 18            | $3s^2 3p^6$              | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 5             | B                        |  <p>Hence, there are 6 single covalent bonds and 2 dative bonds.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |
| 6             | A                        | <p>Option 1 – Correct. Ethylene glycol can form 2 hydrogen bonds per molecule (due to the presence of 2 –OH groups) and ethanol forms 1 hydrogen bond per molecule (due to the presence of 1 –OH group).</p> <p>Option 2 – Correct. Ethylene glycol has a larger electron cloud and so, a larger surface area, resulting in stronger intermolecular instantaneous dipole-induced dipole interactions.</p> <p>Option 3 – Incorrect. Boiling points of simple molecular compounds is dependent on the strength of intermolecular forces of attraction. The strength of covalent bonds does not affect the boiling points of ethylene glycol and ethanol.</p> <p>Hence, only options 1 and 2 are correct.</p> |               |                          |                           |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |    |             |   |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | C |  <p>Hence, coordination number of Na = 6 and coordination number of Cl = 6.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 8  | B | <p>Buffer in blood consists of <math>\text{H}_2\text{CO}_3</math> and <math>\text{HCO}_3^-</math>. To remove the contaminating <math>\text{H}^+</math> ions from the blood, the large reserve of <math>\text{HCO}_3^-</math> removes <math>\text{H}^+</math>.</p> $\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{CO}_3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 9  | C | <p>Total <math>[\text{OH}^-] = (10^{-3} + 10^{-1}) / 2 = 0.0505 \text{ mol dm}^{-3}</math><br/> <math>\text{pOH} = -\lg(0.0505) = 1.297</math><br/> <math>\text{pH} = 14 - 1.297 = 12.7</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 10 | D | <ul style="list-style-type: none"> <li>The formation of a liquid product between <b>J</b> and chlorine indicates that the product is likely to be covalent. Hence, <b>J</b> is a non-metal and unable to conduct electricity.</li> <li>For <b>K</b>, the chloride formed is a solid and dissolves in water to give a solution of pH 7, indicating that the chloride is likely to be sodium chloride. Hence, <b>K</b> is sodium, a metal and so, it is able to conduct electricity.</li> <li>For <b>L</b>, the chloride formed is a solid that dissolves in water to give a solution of pH 6, indicating that the chloride is likely to be magnesium chloride. Hence, <b>L</b> is magnesium, a metal and so, it is able to conduct electricity.</li> </ul> <p>Hence, <b>K</b> and <b>L</b> are good conductors of electricity.</p> |
| 11 | D | <p>Down group 17, the volatility decreases as the boiling points of the elements increases. This is because the size of the electron cloud of the elements increases and the polarisability of the electron cloud increases down the group, leading to an increase in the strength of the intermolecular instantaneous dipole-induced dipole (id-id) interactions. Hence, the energy needed to overcome the id-id interactions increases down the group.</p>                                                                                                                                                                                                                                                                                                                                                                      |
| 12 | A | <p>The nuclear charge increases from <math>\text{F}^-</math> to Ne to <math>\text{Na}^+</math>, and the number of electrons are the same. Hence the attraction between the nucleus and the valence electrons increases from <math>\text{F}^-</math> to Ne to <math>\text{Na}^+</math>, resulting in a decrease in radii from <math>\text{F}^-</math> to Ne to <math>\text{Na}^+</math>.</p> <p>Hence, the radii increases from <math>\text{Na}^+</math> to Ne to <math>\text{F}^-</math>.</p>                                                                                                                                                                                                                                                                                                                                     |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                      |
|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | C | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | No. of moles of $\text{CO}_2 = \frac{300}{24\,000} = 0.0125 \text{ mol}$<br>Volume of $\text{CO}_2 = 0.0125 \times 24\,000 = 300 \text{ cm}^3$       |
|    |   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | No. of moles of $\text{HCl} = 0.0125 \times 2 = 0.025 \text{ mol}$<br>Volume of $\text{HCl} = 0.025 \div 0.50 = 0.05 \text{ dm}^3 = 50 \text{ cm}^3$ |
|    |   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | No. of moles of $\text{NaCl} = 0.0125 \times 2 = 0.025 \text{ mol}$<br>Mass of $\text{NaCl} = 0.025 \times 58.5 = 1.46 \text{ g}$                    |
|    |   | Hence, options 1 and 3 are correct.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                      |
| 14 | C | $A_r = \frac{(63 \times 65) + (65 \times 29)}{94} = 63.6$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                      |
| 15 | B | Oxidation no. of S in $\text{Na}_2\text{S}_2\text{O}_3 = \frac{-[3(-2) + 2(+1)]}{2} = +2$<br>Oxidation no. of S in $\text{SO}_2 = -[2(-2)] = +4$<br>Oxidation no. of S in S = 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                      |
| 16 | C | No. of moles of $\text{FeC}_2\text{O}_4 = 18 \times 10^{-3} \times 0.020 = 3.60 \times 10^{-4} \text{ mol}$<br>No. of moles of $\text{Cr}_2\text{O}_7^{2-}$ needed to oxidise $\text{Fe}^{2+} = \frac{1}{6} \times 3.60 \times 10^{-4} = 6.0 \times 10^{-5} \text{ mol}$<br>No. of moles of $\text{Cr}_2\text{O}_7^{2-}$ needed to oxidise $\text{C}_2\text{O}_4^{2-} = \frac{1}{3} \times 3.60 \times 10^{-4} = 1.2 \times 10^{-4} \text{ mol}$<br>Total no. of moles of $\text{Cr}_2\text{O}_7^{2-} = 6.0 \times 10^{-5} + 1.2 \times 10^{-4} = 1.8 \times 10^{-4} \text{ mol}$<br>Volume of $\text{K}_2\text{Cr}_2\text{O}_7 = 1.8 \times 10^{-4} \div 0.02 = 9.0 \times 10^{-3} \text{ dm}^3 = 9 \text{ cm}^3$ |                                                                                                                                                      |
| 17 | B | heat absorbed by water = $200 \times 4.18 \times 48 = 40128 \text{ J}$<br>heat evolved from combustion = $\frac{40128}{45} \times 100 = 89173 \text{ J}$<br>total energy released per gram of fuel burnt = $89173 \div 1.6 = 55733 = 55700 \text{ J}$                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                      |
| 18 | B | Enthalpy change of formation of water is the enthalpy change when <b>1 mole of <math>\text{H}_2\text{O}(l)</math></b> is formed from its constituent elements at their standard states, i.e., <b><math>\text{H}_2(g)</math> and <math>\text{O}_2(g)</math></b> under standard conditions of 298 K and 1 bar.                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                      |
| 19 | D | option A - Incorrect. Arrow representing $E_a$ should start from the reactants.<br>option B – Incorrect. Arrow representing $\Delta H$ should start from the reactants.<br>option C – Incorrect. Direction of arrow representing $\Delta H$ should be pointing from reactants to products.<br>option D – Correct!                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                      |
| 20 | C | Since the substrate must fit into the active site of the enzyme before catalysis can occur, only properly designed molecules can serve as substrates for a specific enzyme; in many cases, an enzyme will react with only one naturally occurring molecule.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                      |

|    |                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|----|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|----|----|---|----------------------|---|---|---|------------------------|----|-----|----|--------------------------|-----|----|---|-----------------------------------------|--|--|--|---|--|----|----|---|----------------------|---|---|---|------------------------|-----|-----|----|--------------------------|------|----|---|----------------------------------------|--|--|--|---|--|----|---|---|----------------------|---|---|---|------------------------|-----|----|----|--------------------------|------|---|---|------------------------------------|--|--|--|---|--|----|---|----|----------------------|---|---|---|------------------------|----|----------------|----|--------------------------|-----|---------------|---|-----------------------------------------------------|--|--|--|
| 21 | A                                                   | <p>option 1 – Correct. Increasing temperature increases rate of forward and backward reactions.</p> <p>option 2 – Correct. Increasing the temperature will cause the position of equilibrium to shift to the left, favouring the endothermic backward reaction, so as to absorb heat. Hence, yield of methanol decreases.</p> <p>option 3 - Correct. Increasing the pressure will cause the position of equilibrium to shift to the right, favouring the forward reaction that produces less amount of gases, so as to reduce the increased pressure. Hence, the yield of methanol increases.</p> <p>All 3 options are correct.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
| 22 | D                                                   | <table><tr><td rowspan="5">A</td><td></td><td>P</td><td>2Q</td><td>R</td></tr><tr><td>Initial amount / mol</td><td>2</td><td>0</td><td>0</td></tr><tr><td>Change in amount / mol</td><td>-x</td><td>+2x</td><td>+x</td></tr><tr><td>Equilibrium amount / mol</td><td>2-x</td><td>2x</td><td>x</td></tr><tr><td colspan="4">Total no. of moles = 2 + 2x (incorrect)</td></tr></table> <table><tr><td rowspan="5">B</td><td></td><td>2P</td><td>2Q</td><td>R</td></tr><tr><td>Initial amount / mol</td><td>2</td><td>0</td><td>0</td></tr><tr><td>Change in amount / mol</td><td>-2x</td><td>+2x</td><td>+x</td></tr><tr><td>Equilibrium amount / mol</td><td>2-2x</td><td>2x</td><td>x</td></tr><tr><td colspan="4">Total no. of moles = 2 + x (incorrect)</td></tr></table> <table><tr><td rowspan="5">C</td><td></td><td>2P</td><td>Q</td><td>R</td></tr><tr><td>Initial amount / mol</td><td>2</td><td>0</td><td>0</td></tr><tr><td>Change in amount / mol</td><td>-2x</td><td>+x</td><td>+x</td></tr><tr><td>Equilibrium amount / mol</td><td>2-2x</td><td>x</td><td>x</td></tr><tr><td colspan="4">Total no. of moles = 2 (incorrect)</td></tr></table> <table><tr><td rowspan="5">D</td><td></td><td>2P</td><td>Q</td><td>2R</td></tr><tr><td>Initial amount / mol</td><td>2</td><td>0</td><td>0</td></tr><tr><td>Change in amount / mol</td><td>-x</td><td><math>+\frac{x}{2}</math></td><td>+x</td></tr><tr><td>Equilibrium amount / mol</td><td>2-x</td><td><math>\frac{x}{2}</math></td><td>x</td></tr><tr><td colspan="4">Total number of moles = <math>2 + \frac{x}{2}</math> (correct)</td></tr></table> | A              |     | P  | 2Q | R | Initial amount / mol | 2 | 0 | 0 | Change in amount / mol | -x | +2x | +x | Equilibrium amount / mol | 2-x | 2x | x | Total no. of moles = 2 + 2x (incorrect) |  |  |  | B |  | 2P | 2Q | R | Initial amount / mol | 2 | 0 | 0 | Change in amount / mol | -2x | +2x | +x | Equilibrium amount / mol | 2-2x | 2x | x | Total no. of moles = 2 + x (incorrect) |  |  |  | C |  | 2P | Q | R | Initial amount / mol | 2 | 0 | 0 | Change in amount / mol | -2x | +x | +x | Equilibrium amount / mol | 2-2x | x | x | Total no. of moles = 2 (incorrect) |  |  |  | D |  | 2P | Q | 2R | Initial amount / mol | 2 | 0 | 0 | Change in amount / mol | -x | $+\frac{x}{2}$ | +x | Equilibrium amount / mol | 2-x | $\frac{x}{2}$ | x | Total number of moles = $2 + \frac{x}{2}$ (correct) |  |  |  |
| A  |                                                     | P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                | 2Q  | R  |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Initial amount / mol                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                | 0   | 0  |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Change in amount / mol                              | -x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                | +2x | +x |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Equilibrium amount / mol                            | 2-x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | 2x  | x  |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Total no. of moles = 2 + 2x (incorrect)             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
| B  |                                                     | 2P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2Q             | R   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Initial amount / mol                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0              | 0   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Change in amount / mol                              | -2x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | +2x            | +x  |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Equilibrium amount / mol                            | 2-2x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2x             | x   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Total no. of moles = 2 + x (incorrect)              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
| C  |                                                     | 2P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Q              | R   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Initial amount / mol                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0              | 0   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Change in amount / mol                              | -2x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | +x             | +x  |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Equilibrium amount / mol                            | 2-2x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | x              | x   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Total no. of moles = 2 (incorrect)                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
| D  |                                                     | 2P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Q              | 2R  |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Initial amount / mol                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0              | 0   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Change in amount / mol                              | -x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $+\frac{x}{2}$ | +x  |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Equilibrium amount / mol                            | 2-x                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\frac{x}{2}$  | x   |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
|    | Total number of moles = $2 + \frac{x}{2}$ (correct) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
| 23 | C                                                   | <p>Geckos have arrays of millions of microscopic hairs, or setae, on the bottoms of their feet. Each seta ends in an array of <b>nanosstructures</b>, called spatulae, that function as a dry adhesive. The <b>high surface area to volume ratio</b> results in the adhesive property.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |
| 24 | C                                                   | <div></div> <p>Hence, there are 22 hydrogen atoms.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |     |    |    |   |                      |   |   |   |                        |    |     |    |                          |     |    |   |                                         |  |  |  |   |  |    |    |   |                      |   |   |   |                        |     |     |    |                          |      |    |   |                                        |  |  |  |   |  |    |   |   |                      |   |   |   |                        |     |    |    |                          |      |   |   |                                    |  |  |  |   |  |    |   |    |                      |   |   |   |                        |    |                |    |                          |     |               |   |                                                     |  |  |  |

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25 | A | <p><i>Melamine</i> is a thermoset with strong covalent bonds holding the layers together and so, it does not melt. <i>Acrylic</i> is a thermoplastic whereby the chains are not cross-linked and the forces between the chains are weaker than the crosslinks in melamine. Hence, <i>acrylic</i> melts on heating.</p> <p>Option 1 – Incorrect. <i>Melamine</i> cannot be melted so, it cannot be recycled.<br/> Option 2 - Incorrect. <i>Acrylic</i> does not have crosslinks<br/> Option 3 – Correct. As the crosslinks are strong covalent bonds, heating does not break these crosslinks and so, <i>melamine</i> does not melt.</p>                                              |
| 26 | C | Rate of biodegradation of PVC is slow. PVC burns to produce $\text{CO}_2$ , $\text{H}_2\text{O}$ and $\text{HCl}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 27 | C | The esters are $\text{HCO}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ , $\text{HCO}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$ , $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$ , $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ , $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_3$ , $(\text{CH}_3)_2\text{CHCO}_2\text{CH}_3$ . Hence, there are a total of 6 esters.                                                                                                                                                                                                                                                                                |
| 28 | D |  <p style="text-align: center;">coenzyme <math>\text{Q}_{10}</math></p> <p>There are <math>n</math> <math>\pi</math> bonds in the brackets. There are 4 <math>\pi</math> bonds outside the brackets. Hence, there is a total of <math>n + 4</math> <math>\pi</math> bonds in one molecule.</p>                                                                                                                                                                                                                                                                                                     |
| 29 | C | <p>The functional group in delta lactone is ester which is formed from an alcohol and carboxylic acid.</p> <p>option A – incorrect. The functional groups present are alcohol and ketone.<br/> option B – incorrect. The functional groups present are alcohol and aldehyde.<br/> option C – correct. The functional groups present are alcohol and carboxylic acid.<br/> option D – correct. Both functional groups present are alcohols.</p>                                                                                                                                                                                                                                       |
| 30 | A | <p>2-hydroxypentanenitrile is <math>\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CN}</math>.</p> <p><math>\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CN}</math> is formed from <math>\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}</math>, via addition reaction using <math>\text{HCN}</math> with <math>\text{NaCN}</math> catalyst, cold.</p> <p>The intermediate <math>\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}</math> is formed from <math>\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}</math> (butan-1-ol), via oxidation reaction using acidified <math>\text{K}_2\text{Cr}_2\text{O}_7</math>, heat with immediate distillation.</p> |